MILITARY SPECIFICATION

FLASK, SEPARATOR, 3000 PSI, K-MONEL

1. SCOPE

1.1 SCOPE. This specification covers K-Monel high pressure flasks with Monel end plugs for separation of impurities from oxygen in submarine oxygen generator systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (K-Monel)

MILITARY
MIL-T-1368 - Tubing, Nickel-Copper Alloy, Seamless and Welded
MIL-E-22200/3 - Electrodes, Welding, Covered: Nickel Base Alloy: and Cobalt Base Alloy

STANDARDS

FEDERAL
FED-STD-151 - Metals: Test Methods

MILITARY
MIL-STD-271 - Non-Destructive Testing Requirements for Metals

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. -

3.1.1 Flasks. - The material for the flasks shall be seamless K-monel tube having chemical composition in accordance with class A of QQ-N-286. The material shall be hot finished, annealed and age hardened, and shall have the mechanical properties specified in table I.

Table I - Mechanical properties.

Yield strength, at 0.2 percent offset	90,000 psi minimum
Tensile strength	130,000 psi minimum
Elongation	20.0 percent in 2 inches

3.1.2 End plugs. - End plugs shall be nickel-copper alloy in accordance with class A of QQ-N-281.

3.2 Dimensional characteristics. - The dimensional characteristics of flasks shall be as shown in table II. Unless otherwise specified herein, the difference between the major and minor diameters shall not exceed 2 percent of the flask outside diameter.

Table II - Dimensional characteristics.

Capacity	0.5 cubic feet
Outside diameter $\pm$ 3/32 inch	6.625 inches
Length + 2 inches - 0 inches	3 feet 10 inches
Wall thickness, minimum	0.375 inches
Hydrostatic test pressure	5000 psi
Applicable figure	1

3.3 Fabrication. - The body of the flask shall be in the form of a straight cylinder as shown on figure 1. Each end shall be hot formed to provide a neck for the end connections (see figure 2). Flasks shall be formed of seamless tubular material.

3.3.1 Annealing. - After completion of forging operations, flasks shall be annealed, then rapidly quenched from the annealing temperature.

3.3.2 Age hardening. - After finish machining, flasks shall be cleaned of all machining lubricants and age hardened.

3.3.3 Neck threads. - The neck threads shall be Unified National threads in accordance with Handbook H-28 (see figure 2).

3.4 Cleaning. - Flasks shall be thoroughly cleaned of all loose particles and other contamination by abrasive blast, if necessary, followed by a wash with an acid pickle. The flask shall be thoroughly dried and immediately subjected to the oil free test of 4.4.3.4. All oils that come in contact with a flask under production shall be of a type that will fluoresce under an ultra-violet light examination.

3.5 Assembly. - Each flask shall be furnished with an inlet and outlet plug, and a drain plug. (See figures 3 and 4.) End plugs shall be torqued to a minimum of 1000 foot pounds, and seal welded (see figure 5). Welding shall be done by the shielded metal arc method using Monel welding electrode conforming to type 8N10 of MIL-E-22200/3. The inlet/outlet plug shall be assembled with inlet pipe and deflector as shown in figures 3 and 6 prior to installation in flask (figure 1).

3.6 Label plate. - A label plate shall be made for each flask of 1/16 inch thick aluminum. The label plate shall be attached to the neck of the flask by suitable metal bands. The following information shall be stamped on the plate in 1/4 inch letters:

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a. Manufacturer's name and identification number
b. Weight to nearest pound
c. Operating pressure - 3000 psi
d. Material - QQ-N-286
e. Contract number
f. Specification number
g. Date

3.7 Indentations on Flask. - Indentations, other than those required by a manufacturer for identification of a flask during fabrication, are not permitted on the surface of the flask. Such markings may appear on the neck portion only. They shall be made with a round bottom stamp and shall be not over 1/16 inch deep (see figure 4).

3.8 Workmanship. - Flasks shall be free from cracks, and other injurious defects. Threads shall be smooth, properly formed and free from burrs, nicks and gouges. The label plate shall be clear and legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - A lot shall consist of all flasks from the same heat or melt, and heat treated in the same heat treatment furnace charge.

4.2.2 Test sample. - A test sample for tensile and yield tests (see 4.4.2.1) shall be furnished with each lot of flasks. The test sample shall be in the shape of a right circular cylinder having the same inside diameter and wall thickness as the flasks which it represents. The sample shall be of the same heat of K-monel, and shall be subjected to the same manufacturing processes (except swaging) and heat treatment as the flasks which it represents. The methods of quenching the test sample must duplicate that of the flasks.

4.3 Visual and dimensional examination. - Each flask shall be visually and dimensionally examined to verify compliance with this specification. Flasks shall be clean and free from loose scale, chips, or foreign materials. Any flask containing visual or dimensional defects shall not be offered for delivery.

4.4 Test procedure. -

4.4.1 Chemical analysis. - Prior to flask fabrication a suitable sample shall be taken from the pipe, billets or plate of each heat of K-monel to be used for flask fabrication, and shall be subjected to a chemical analysis to determine compliance with 3.1.1. Sample selection and testing shall conform to FED-STD-151. Failure of the chemical analysis to conform as specified shall be cause for rejection of the heat of material represented.

4.4.2 Destructive tests. - The tests in 4.4.2.1 require test specimens cut from the central 12 inch section of the test sample (see 4.2.2).

4.4.2.1 Tensile and yield. - Four tensile specimens shall be cut longitudinally from the specified portion of the test sample (see 4.2.2), spaced 90 degrees apart and shall be tested in accordance with method 211.1 of FED-STD-151 to determine compliance of tensile, yield and elongation properties specified in table I. Failure to pass this test shall be cause for rejection of the lot represented.

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4.4.3 Flask tests. -

4.4.3.1 Hydrostatic test. - After assembly of end plugs, each flask shall be tested in a water jacket to a test pressure equal to 5/3 (five thirds) the design working pressure of the flask. Cold tap water shall be used for this test. Neither the water nor the flask shall be heated. The test pressure shall be held at least 30 minutes. As a result of this test the flask shall not rupture or show a permanent set exceeding 10 percent of the total volumetric expansion. There shall be no evidence of leakage through any portion of the flask or end plug. Any flask that fails to comply with this test shall not be offered for delivery.

4.4.3.2 Ultrasonic inspection. - After the hydrostatic tests have been completed, the cylindrical portion of each flask shall be 100 percent ultrasonically inspected in accordance with MIL-STD-271. The ultrasonic inspection equipment shall be calibrated to detect a notch equal to 3 percent of the flask wall thickness. A 100 percent loss of back reflection indicates a discontinuity size equal to or greater than the 3 percent calibration notch. Any flask showing such a discontinuity size shall not be offered for delivery. All flasks having discontinuities producing a loss in back reflection between 50 percent and 100 percent that are greater than 3 inches in length shall be reported to the bureau or agency concerned together with all other discontinuities producing a loss in back reflection greater than 25 percent. The report shall be submitted in sketch form orientating all reportable discontinuities, longitudinally or circumferentially, as found, and indicating their severity and length. If the reported discontinuities are lineally disposed along the flask and the total length of all discontinuities in 12 inches of flask length is 6 inches, the flask shall not be offered for delivery unless the discontinuities are located on the surface and can be repaired without violating the specified minimum wall thickness. Repairs shall be by grinding, unless otherwise approved by the bureau or agency concerned.

4.4.3.2.1 Ultrasonic inspection equipment shall be recalibrated periodically to insure optimum results. It shall also be recalibrated if there is any change in frequency or crystal size used.

4.4.3.3 Liquid penetrant inspection. End sections shall be inspected by liquid penetrant in accordance with MIL-STD-271. All linear or linearly disposed circular indications as well as indications found in areas where abrupt changes in direction occur shall be treated as defects. Such defects shall be cause for rejection if they are greater than .0625 inch in depth. If less than .0625 inch deep, they may be faired into the adjacent area at a ratio of four to one by grinding.

4.4.3.4 Oil-free test. Flasks, after cleaning as specified in 3.4, shall be inspected internally by ultra-violet light ("Black light": 3200 to 4000 Angstrom units). If there is any evidence of fluorescence on the interior or machined surfaces, the flask shall be recleaned and resubmitted to the oil-free test.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Packaging. Flasks shall be cleaned of all external oil, or dirt, sealed in a polyethylene bag, to insure cleanliness, and packaged in a manner sufficient to afford protection against physical damage during shipment from the supply source to the using activity and until early installation. Openings shall be closed by caps. (See figure 7.)

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage from handling during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. Shipment marking information shall be provided in accordance with the supplier's commercial practice. The information shall include nomenclature, contract or order number, manufacturer's name and destination. Securely attached to each flask shall be a waterproof tag indelibly marked with all pertinent shipping information.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.

Preparing activity:
Navy - Ships
(Project 8120-N017Sh)

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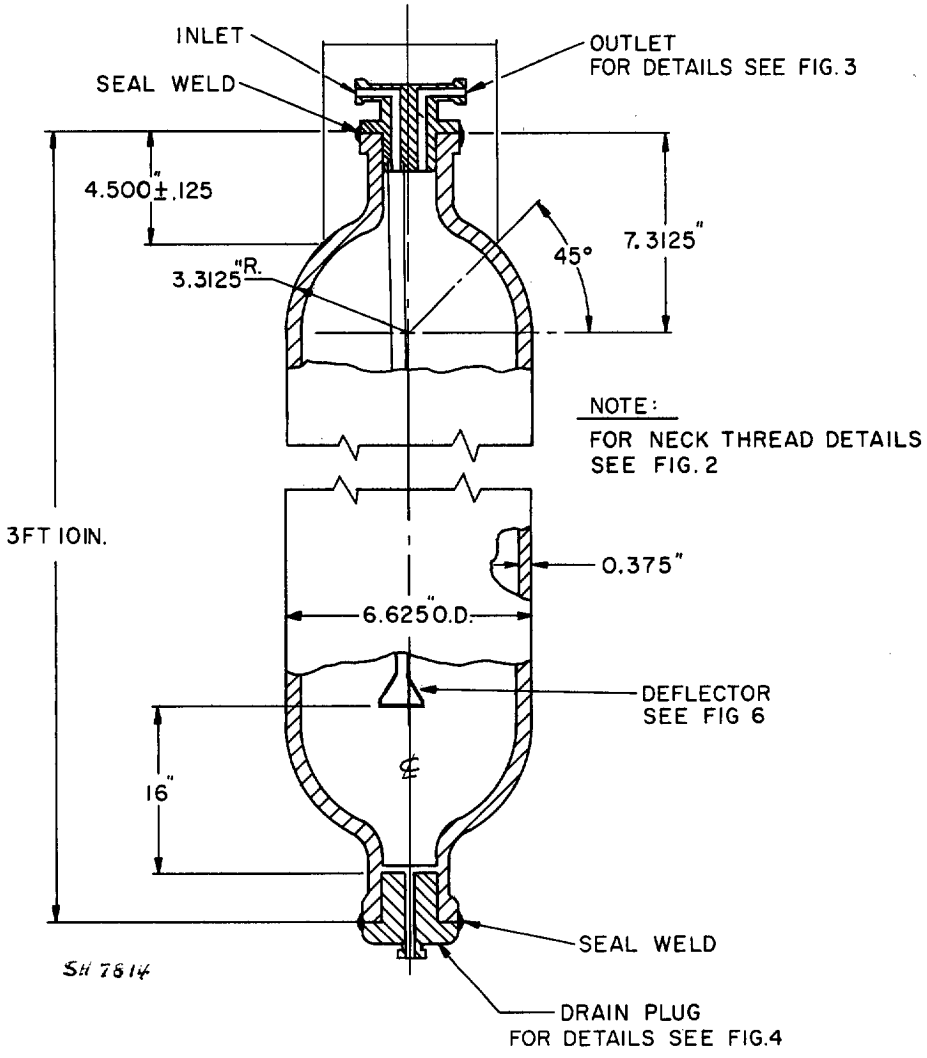
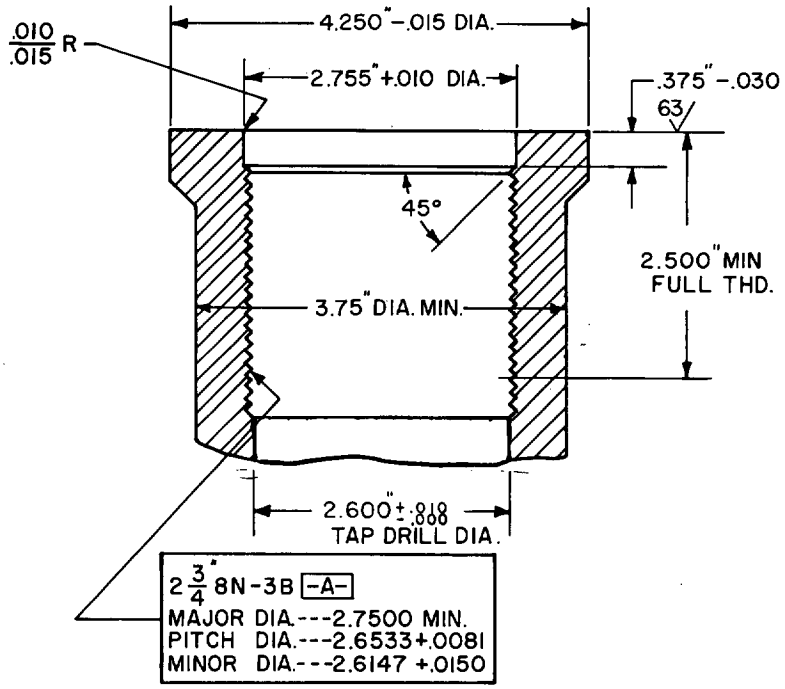


FIGURE 1 FLASK ASSEMBLY

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SH 7815 NOTE:

SURFACE -B- MUST BE PERPENDICULAR
TO AXIS -A- WITHIN .006 T.I.R.

FIGURE 2 NECK THREAD DETAILS

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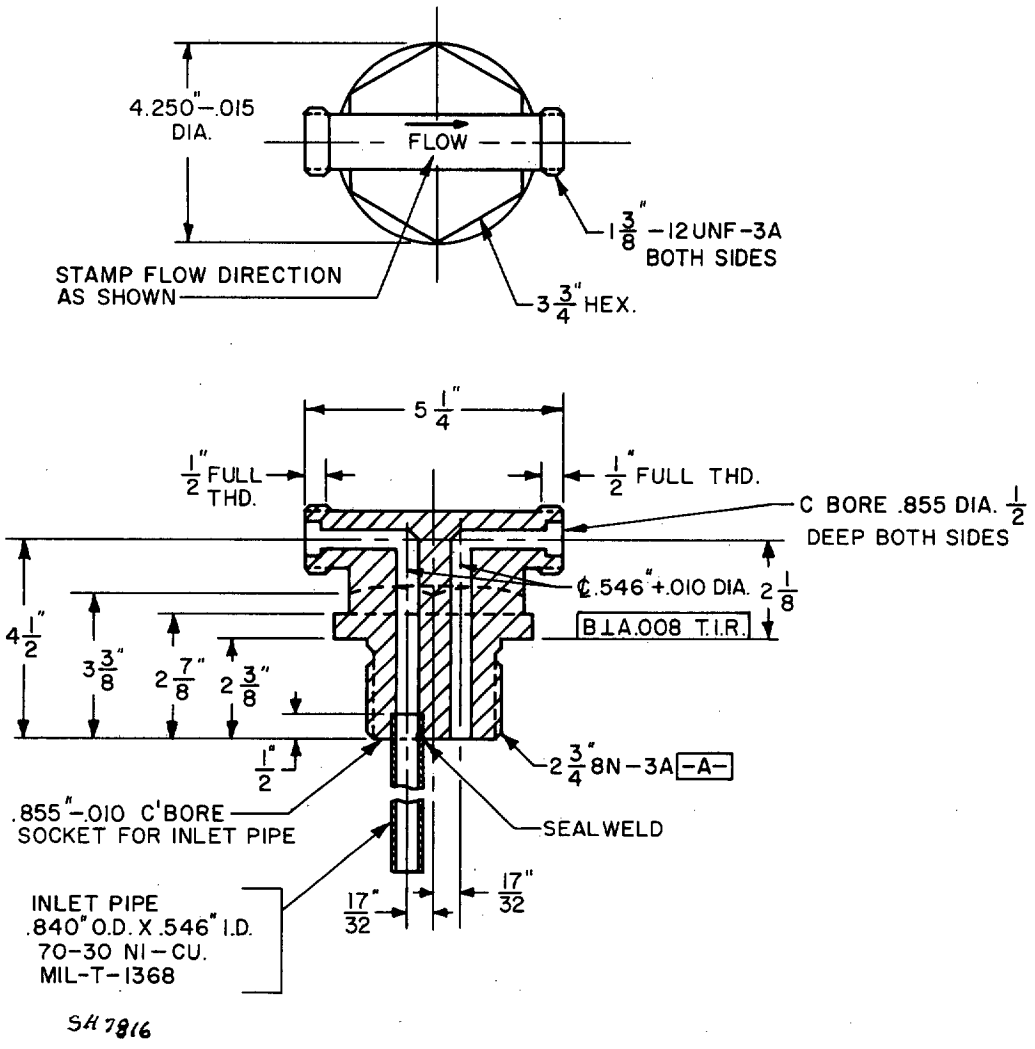
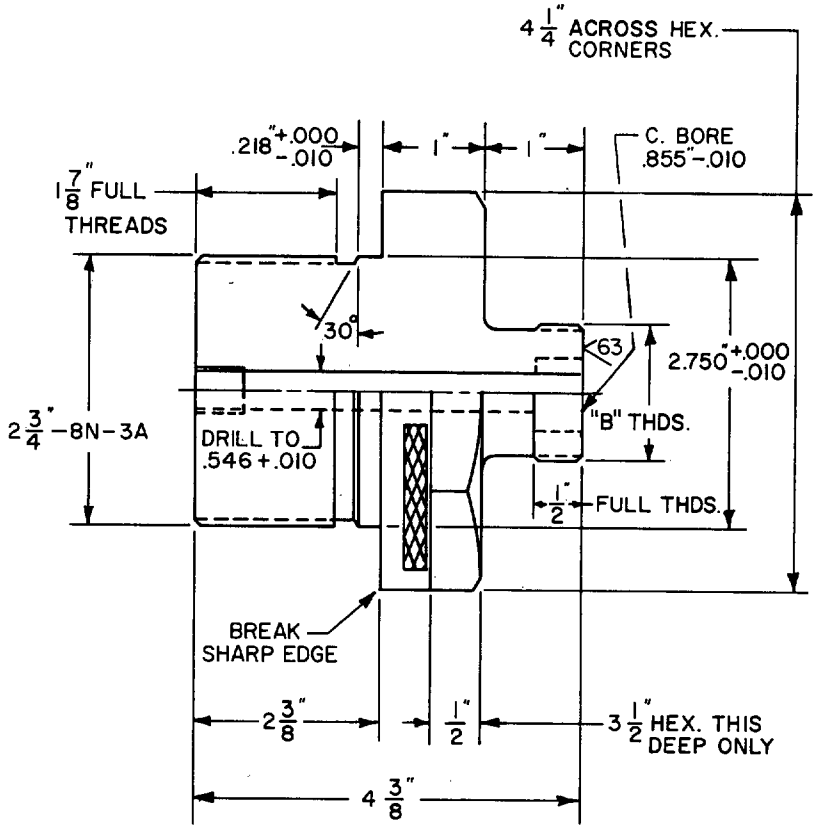


FIGURE 3 – INLET AND OUTLET PLUG

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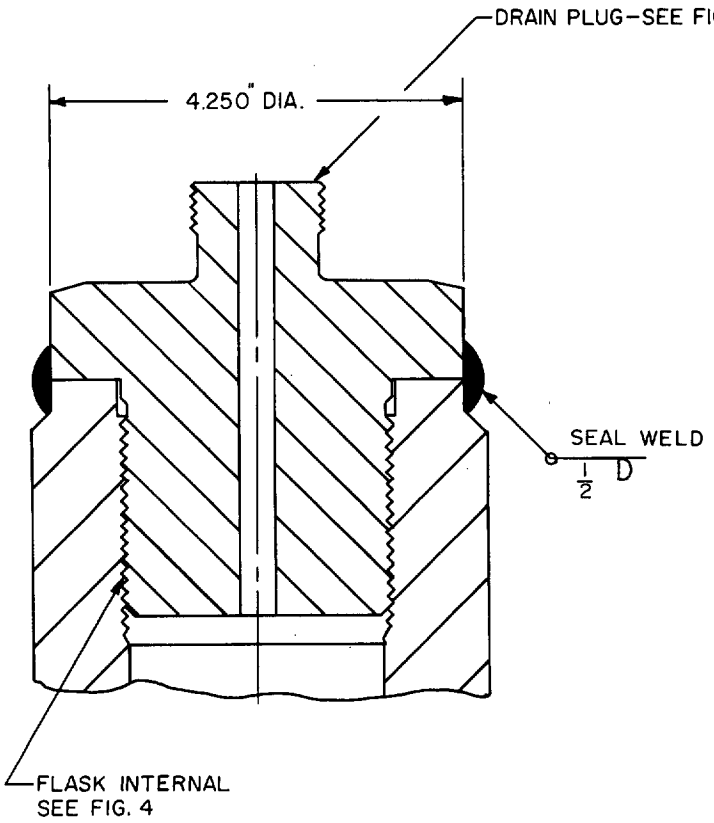


PLUG TYPE	A	B	NOTE
DRAIN		1 3/8"-12 UNF-3A	AT BOTTOM OF FLASK

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FIGURE 4 - PLUGS FOR FLASK ASSEMBLY

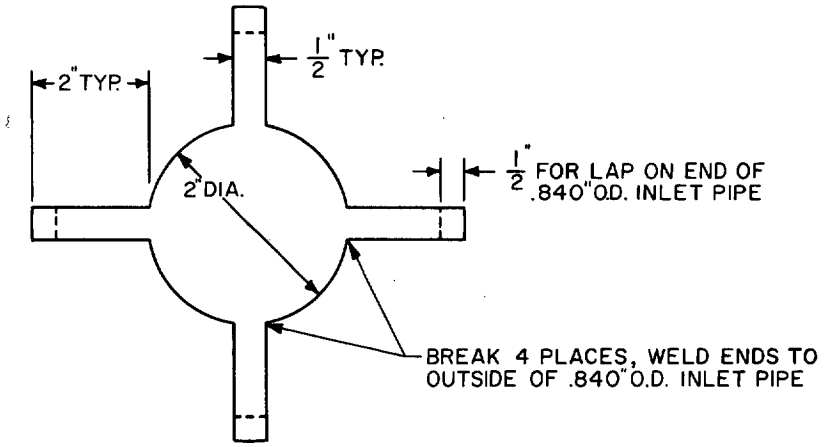
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FIGURE 5– TYPICAL PLUG ASSEMBLY

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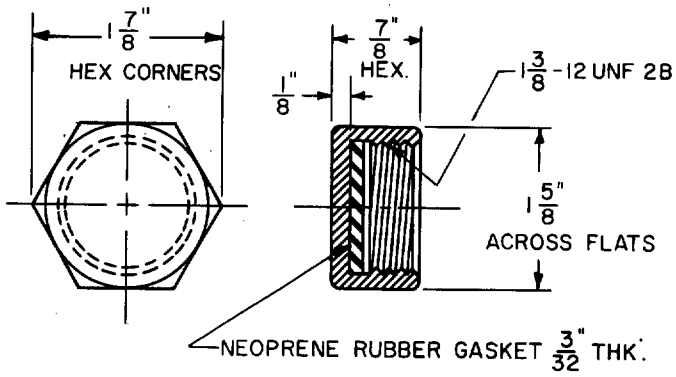


MATERIAL – 20 GA. SHEET, QQ-N-281

SH 7819

FIGURE 6 - DEFLECTOR BLANK

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CAP FOR INLET & OUTLET PLUG FIG. 3 AND
DRAIN PLUG FIG. 4
MATERIAL- COMMERCIAL BRASS

SH 7820

FIGURE 7 - SEAL CAP

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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